Out-of-bag error and variable importance for random forests

Breiman's own measures of a forest, tested against his Table 2 on 19 data sets.

Tested against	Breiman (2001), *Random forests*, *Machine Learning*: Table 2 (p. 11) and Section 10 (pp. 19–21). 14 real data sets from PollisDatasets, four simulated as Breiman did, and the zip-code digits of Hastie, Tibshirani and Friedman (2009) (downloaded only if the user agrees).
Missing in Julia	the out-of-bag error and permutation importance. DecisionTree.jl grows forests, but has neither.
Notebook	**Compose** › **New Methods** › **Browse Methods**: *Out-of-bag error and variable importance for random forests* (`oob.ipynb`)
Result	34 of 38 Selection and Single input errors within simulation noise of Table 2; importance picks the inputs Breiman's did.

The problem

A random forest grows each tree on a bootstrap sample, so about a third of the cases are left out of each tree. Breiman (2001) used these out-of-bag cases twice: to estimate the forest's error without a test set, and to measure how much the forest relies on each input. Both are cited constantly and computed differently by different libraries. Breiman's values are averages of 100 random runs, so the test is statistical: a Julia average, with its standard error, against his.

The method

Forest-RI and the out-of-bag error

Forest-RI (p. 9) grows each tree on a bootstrap sample of the training cases, splitting at each node on the best of F inputs drawn at random, by the Gini criterion of Breiman, Friedman, Olshen and Stone (1984), unpruned; the forest votes. Run down only the trees that did not see them, the out-of-bag cases give the forest's **out-of-bag error** (Section 3.1), and the error of each tree alone, Table 2's "One tree". `forest` returns the trees and both errors; `classify` runs cases down a tree without recursion, since the deepest trees overflow the stack of DecisionTree.jl's recursive `apply_tree`.

Permutation importance

Permute input m among the out-of-bag cases of each tree and run them down again: the percent increase in the out-of-bag error measures how much the forest relies on m (Section 10).

What the tests show

Breiman grew two forests per data set, with $F = 1$ and $F = \mathrm{int}(\log_2 M + 1)$, and took the test error of the one with the lower *out-of-bag* error: the **Selection** column. **Single input** is the test error with $F = 1$. A value is *within noise* when the two averages differ by less than two standard errors of a difference ($\sqrt{2}$ s.e.).

Test	What is compared	Within noise
Table 2, Selection	test error of the forest chosen by out-of-bag error	17 of 19
Table 2, Single input	test error of the forest with $F = 1$	17 of 19
Table 2, One tree	out-of-bag error of single trees	7 of 19
Section 10	errors of forests on the inputs importance picks	4 of 5

Where they differ

Liver Selection 27.3 vs 25.1 (2.0 s.e.); twonorm Single input 3.7 vs 3.9 (2.1 s.e.); ringnorm Selection 4.4 vs 4.9 and Single input 4.3 vs 4.9 (2.4 and 4.7 s.e.). Both ringnorm values are lower than Breiman's.

One tree: most differ by about one point (pima 31.9 vs 33.1, liver 39.3 vs 40.6), and vowel by more, 36.1 vs 30.4. The test is strict there: each value averages 100 trees over 100 runs, so the standard errors are small, and small differences between DecisionTree.jl's tree code and Breiman's (tie-breaking, split points) show as several standard errors.

Importance picks the same inputs

On diabetes, importance ranks input 2 (plasma glucose) first, as Breiman's Figure 4 does. Forests on the inputs it picks give 29.9 / 24.0 / 30.2 (input 2 / all eight / 2 and 8) against 29.7 / 23.1 / 29.4, within noise, and 28.1 vs 26.4 for inputs 2 and 6 (2.4 s.e.). On the 1984 House votes, input 4 (freezing physicians' fees) comes first, as in Figure 6, and a forest on it alone

errs 4.7% against 4.3%, within noise.

Still open

- **Ringnorm and twonorm.** The forests are below Breiman on these two simulated sets, and arc-fs (its own document) shows the same small gap on the same two sets. A gap shared by two methods points to how the trees are grown, or to the generators, which follow the description in Breiman (1998), p. 7. Not explained.
- **One tree**, about one point off on most data sets: a difference in the single trees, not in the forests' votes.

Further considerations

- **Replicating random results.** Every run has its own seed, so the results do not depend on the number of threads, and every comparison carries its standard error: a disagreement is a number of standard errors, not a judgement.
- **Upstream.** Out-of-bag error and permutation importance are small additions to DecisionTree.jl, and Table 2 is a ready-made benchmark for them.
- **Next.** Breiman's third use of out-of-bag cases, proximities (for outliers and missing values), and his regression forests (Section 11), each tested against his tables.

References

Breiman, L. (1998). Arcing classifiers. *The Annals of Statistics* 26(3), 801–849. doi:10.1214/aos/1024691079

Breiman, L. (2001). Random forests. *Machine Learning* 45(1), 5–32. doi:10.1023/A:1010933404324

Breiman, L., Friedman, J. H., Olshen, R. A. and Stone, C. J. (1984). *Classification and Regression Trees*. Wadsworth.

Hastie, T., Tibshirani, R. and Friedman, J. (2009). *The Elements of Statistical Learning*, 2nd ed. Springer: the zip-code data.

Methods written and tested with Claude Code (Anthropic) in Pollis, from the published papers and their data.

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